Literature Review: Deep Learning for Seizure Detection and Prediction from Non-EEG Wearable Signals (2013–2026)

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Contents

Abstract

Background

Wearable devices using non-EEG biosignals offer potential for continuous ambulatory seizure monitoring without the burden of scalp electrodes. This systematic review examines the current state of deep learning approaches for seizure detection and forecasting using accelerometry, cardiac signals, and other wearable modalities.

Methods

Following PRISMA guidelines, 13 primary studies (9 detection, 4 forecasting) published between 2013 and 2026 were reviewed. Studies were evaluated following the framwork proposed by Webster and Watson, 2002 based on the follwing metrics: sensitivity, false alarm rate (FAR), validation methodology, and ILAE phase of development.

Results

Detection of convulsive seizures achieves high sensitivity (71.6-100%) using primarily accelerometry-based approaches. Only two systems approach the ILAE Phase 3 benchmark for validated devices: Fine et al. (tonic seizures, 100% sensitivity, 0.023/h FAR) and Dong et al. (nocturnal severe seizures, 71.6% sensitivity, 0.165/h FAR). ECG-based detection consistently fails clinical utility due to excessive FAR (1.91-39.75/h). Nighttime monitoring substantially outperforms daytime (1/61 nights vs. 1/9 days).
Seizure forecasting remains immature. When evaluated with rigorous leave-one-subject-out cross-validation, only 43.5% of patients achieve better-than-chance performance. Patient-specific tuning inflates performance by 30-40 percentage points, creating unrealistic expectations for real-world deployment. Circadian patterns provide the most reliable forecasting signals, but wearable-only features fail at daily resolution.
Validation rigor is insufficient: only 3 of 13 studies employ prospective designs, and leave-one-subject-out cross-validation remains rare. Eleven of 13 studies validate in controlled epilepsy monitoring unit settings, likely overestimating real-world performance.

Conclusions

Wearable seizure detection for convulsive seizures has achieved clinical viability for specific use cases, particularly nocturnal monitoring. However, seizure forecasting faces a fundamental responder problem that prevents clinical translation. False alarm rates remain the primary barrier for daytime detection, and real-world prospective validation is urgently needed. Future research must prioritize standardized metrics, edge deployment optimization, and identification of baseline biomarkers to predict which patients will benefit from wearable monitoring systems.

1 Introduction

1.1 Problem Statement and Motivation

Epilepsy, a chronic neurological disorder affecting 60 million people wordlwide, with approximately one-third of patients remaining drug-resistant (Hussein et al., 2018). Sveins-

son et al., 2020 found that 69% of Deaths due to a Generalized tonic-clonic seizure could have been prevented if not left unattended, highlighting the demand for reliable and timely prediction.
While reliable prediction is already possible through electroencephalography (EEG), its operational complexity render it unsuitable for ambulatory use. To address the need for ambulatory detection many are experimenting with utilizing autonomic and motion signals to detect and predict seizures.
Translating and analysing these often noisy and complex time-series physiological signals has pushed traditional machine learning (ML) approaches to their limits. These approaches require manual engineering such as cleaning and extracting of the data, often failing to capture the intricate and non-linear complex temporal dynamic of these signals. It is this limitation that deep learning (DL) models are designed to overcome, with their ability of hierarchical and learned feature extraction uniquely addressing these limitations. This review therefore examines and assesses recent advances in time-series deep learning - particularly convolutional neural networks (CNNs) and long short term memory (LSTMs) - applied to seizure detection and prediction using non-EEG wearable biosignals. Central to this analysis is the question of how architectural choices, training paradigms (e.g., transfer learning, personalization), and thorough prospective evaluation can converge to meet the stringent clinical requirements of high sensitivity and minimal false alarm rates (FAR) in real-world deployment.

1.2 Key Research Question

How do different deep learning architectures and biosignal modalities excluding EEG compare in their ability to achieve a optimal trade-off between sensitivity and false alarm rate for ambulatory seizure monitoring?

2 Theoretical Foundations for Ambulatory Seizure Monitoring

2.1 The Ambulatory Monitoring Paradigm: Requirements and Challenges

Beniczky et al., 2021, Chen et al., 2022 found that translation of seizure detection from the controlled clinical environment to an ambulatory setting redefines approaches and performance requirements. In clinical environments maximum sensitivity is prioritized over FAR, when engineering for ambulatory use a balance of both metrics is needed. Failing to detect a seizure (low sensitivity) poses a safety risk, While excessive false alarms (high FAR) leads to desensitizing and "alarm fatigue" and the device being ignored or not used at all. Users reported that devices should achieve high sensitivity ($\geq$90%) with low false alarm rates — acceptable FARs were 1-2/week for patients with $\geq$1 seizure/week and 1-2/month for patients with <1 seizure/week (Sivathamboo et al., 2022). While more complex models can find more intricate patterns, they also introduce the "black-box" problem wherein the model lacks transparency AbuAlrob et al., 2025; Ghaderi, 2025. The design should incorporate explanations for decisions to ensure user trust and regulatory compliance AbuAlrob et al., 2025; Miron et al., 2025.

2.2 Physiological Biosignals for Wearable Sensing

Wearable devices circumvent the impracticality of EEG by capturing peripheral biosignals that serve as proxies for central nervous system activity during seizures. These proxies will inherently relay incomplete and noisy signals so a combination of multiple sensors is required. The acceleration/attitude sensor excels at detecting muscle convulsions but is noisy in daily use, as it captures all movement. The electrodermal activity (EDA) sensor detects autonomic arousal by measuring the skin conductance associated with sweat gland activity.

2.3 Seizure Manifestations and Corresponding Biosignals

Wearable biosignals act as peripheral proxies for cerebral activity during seizures. Different seizure types produce distinct physiological signatures across these signals, informing detection approaches. This relationship centers on two primary manifestations: motor activity and autonomic activation.

Convulsive seizures, such as generalized tonic-clonic events, produce clear motor signatures captured by accelerometry (ACC), which measures movement and posture changes characteristic of ictal events (Wang et al., 2025; Wu et al., 2024).

At the same time, seizures drive autonomic responses recorded through complementary biosignals. Electrodermal activity (EDA) measures sympathetic nervous system arousal via skin conductance changes (Jain et al., 2017; Miron et al., 2025). Cardiac activity provides particularly strong autonomic proxies: electrocardiography (ECG) captures heart rate and rhythm alterations (Ghaderi, 2025; Leal et al., 2017), while heart rate variability (HRV) quantifies autonomic balance dynamics (Mason et al., 2024; Pavei et al., 2017).

In wearable applications, photoplethysmography (PPG) serves as a practical alternative for cardiac monitoring, deriving pulse rate and variability though with greater motion sensitivity than ECG (Chen et al., 2022; Seth et al., 2023).

As a result, detection strategies combine these biosignals according to target seizure types, with multimodal fusion providing the most reliable approach for ambulatory monitoring.

2.4 Evaluation Frameworks: From Retrospective Analysis to Prospective Validity

The clinical validity of a seizure detection algorithm is determined by the rigor of its evaluation, formalized in the ILAE's phased framework (Phases 0-4) (Beniczky & Ryvlin, 2018; Beniczky et al., 2021). This framework classifies studies into five phases: Phase 0 (proof of concept), Phase 1 (retrospective, small sample), Phase 2 (minimum 10 patients with seizures, 15 recorded seizures), Phase 3 (prospective, multicenter, $\geq$30 seizures from $\geq$20 patients, real-time detection), and Phase 4 (real-world home validation).

Retrospective studies (Phases 0-2), while foundational, risk substantial data leakage and optimistic bias if they use non-chronological data splits, allowing future information to inform training (Kalousios et al., 2024; Wong et al., 2023). Pseudo-prospective validation addresses this by enforcing chronological order, training only on past seizures to test on subsequent ones, providing a more realistic performance estimate (Kalousios et al., 2024; Lu et al., 2025).

The clinical gold standard is a prospective trial (Phase 3), which requires a locked algorithm, a video-EEG reference standard, and testing on new, unseen data from multiple centers. The ILAE systematic review identified only three Phase 3 studies meeting these strict validation standards (Beniczky et al., 2021). These demonstrated 90-96% sensitivity for generalized tonic-clonic seizures (GTCS) and focal-to-bilateral tonic-clonic seizures (FBTCS) with false alarm rates of 0.2-0.67 per 24 hours (approximately 0.008-0.028/h). This represents the clinically validated performance benchmark for convulsive seizure detection.

What constitutes an acceptable false alarm rate depends on clinical context and user perspective. Systematic surveys reveal important differences in tolerance: patients and caregivers typically require 100% sensitivity and accept approximately one false alarm per seizure (or one per week for seizure-free patients). In contrast, clinicians consider 90% sensitivity adequate and tolerate between two false alarms per week and one per month, depending on seizure frequency. This gap between user and clinician expectations creates a fundamental tension for device development.

The final step is in-field evaluation (Phase 4) in patients' homes, where metrics like the false alarm rate per hour (FAR/h) become decisive for practical usability, as demonstrated by long-term studies like Nightwatch (Miron et al., 2025; Shum & Friedman, 2021).

Currently, the ILAE recommends wearable devices only for GTCS/FBTCS detection in unsupervised patients where alarms can enable rapid intervention - a weak/conditional recommendation based on moderate evidence. For all other seizure types, the ILAE does not recommend clinical use of currently available wearable devices (Beniczky et al., 2021).

3 Methodological Approach and Structure

The review follows structured IS/medical literature guidance (concept matrix inspired by (Webster & Watson, 2002) and PRISMA principles (Tugwell & Tovey, 2021)). The methodological approach uses a multi-stage search and transparent extraction and synthesis procedures as outlined below.

1. **Search strategy (Multi-stage, 2015-2025):**
   - **Scoping:** Google Scholar for broad coverage and identifying candidate papers.
   - **Formal queries:** IEEE Xplore, PubMed, and Scopus using controlled keyword strings and deduplication.
   - **Citation chasing:** Backward and forward citation tracking on key papers to ensure completeness.
2. **Inclusion / Exclusion criteria:**
   - **Inclusion:** Studies using wearable/non-EEG autonomic signals for seizure detection or preictal prediction.
   - **Exclusion:** EEG-only studies, invasive recordings, or papers limited to purely algorithmic simulations without human data.

Extracted dimensions populate two concept matrix tables with the following fields:

1. **Study Matrix (Table 1 for detection, Table 2 for forecasting):**
   - Study identification (author, year)
   - Objective (detection vs forecasting, seizure type)

- Design (prospective, retrospective, pseudo-prospective, LOSO CV)
- Sample (participant count, seizure count)
- Device and sensor location
- Algorithm architecture
- Performance metrics (sensitivity, specificity, FPR)
- Key limitations

2. **Detailed Metrics Matrix (Table 3 for detection, Table 4 for forecasting):**
   - Validation approach (independent test, cross-validation, LOSO)
   - Detection latency
   - Patient-level success rate
   - Precision/PPV
   - Additional clinical metrics (HMS, IoC, AUC, F1)
   - Clinical notes

Additional Extracted Dimensions: The following dimensions were extracted and analyzed to inform the narrative synthesis and dimensional analysis presented in Section 5, complementing the tabular comparison:

- Setting (clinical vs ambulatory)
- Modality details (ECG/HRV, PPG, EDA, ACC)
- Evaluation granularity (sample- vs alarm-based)
- Windowing specifics (window length, stride, overlap)
- Personalization strategies
- Interpretability and safety considerations
- Data characteristics and dataset identity
- Missing data handling (imputation, exclusion) and synthetic data use
- Inference requirements (latency, computational load, hardware)

This two-table approach follows Webster & Watson's concept matrix methodology (Webster & Watson, 2002), balancing comprehensive data extraction with readable summary tables while enabling detailed dimensional analysis in the narrative synthesis.
Prioritization Criteria: Studies including splits preserving patient-specific data, FAR/h reporting and external or cross validation will be prioritized in the synthesis. A PRISMA-style flow diagram will summarize and visualize screening and inclusion.

4 Search Strategy

4.1 PICO-Based Search Methodology

The literature search was structured using the PICO framework (Patient/Problem, Intervention, Comparison, Outcome) to identify relevant studies on deep learning for non-EEG epilepsy detection. The following key terms were used in combination with Boolean operators (AND, OR, NOT):

1. **P (Patient/Problem):** Epileptic seizures in need of detection or prediction.
   - **Keywords:** 'epilepsy', 'seizure', 'ictal', 'preictal'
   - **MeSH Terms (PubMed):** Epilepsy, Se54izures
2. **I (Intervention):** The diagnostic method based on wearable biosignals and deep learning.
   - **Biosignal Keywords:** 'electrocardiogram', 'ECG', 'heart rate variability', 'HRV', 'photoplethysmography', 'PPG', 'electrodermal activity', 'EDA', 'accelerometer', 'ACC'
   - **Technology Keywords:** 'wearable', 'wearable device'
   - **Method Keywords:** 'deep learning', 'machine learning', 'convolutional neural network', 'CNN', 'long short-term memory', 'LSTM', 'neural network'
   - **MeSH Terms (PubMed):** Wearable Electronic Devices, Electrocardiography, Deep Learning
3. **C (Comparison):** The explicit exclusion of studies focused solely on electroencephalography (EEG) as the primary modality.
   - **Exclusion Keywords:** 'EEG', 'electroencephalography', 'iEEG', 'intracranial'
   - **Search Logic:** The `NOT` operator was applied to this group in all database searches.
4. **O (Outcome):** Metrics quantifying the performance of a detection or prediction system.
   - **Keywords:** 'detection', 'prediction', 'sensitivity', 'specificity', 'false alarm rate', 'FAR', 'area under the curve', 'AUC'

4.2 Exploration

Due to the insufficient amount of papers gathered from systematic review an exploration was done on the already existing papers. The exploration consisted of graph tools to visualize backlinks and gemini deep research feature to find relevant papers. Through this method an additional 4 papers were aquired bringing the total number of papers up to 13 and expanding the time frame to 2013-2026.

4.3 Practical Filters and Reproducibility

Consistent with the overall review scope, searches were restricted to **2015-2025** and **journal articles** with few exceptions made regarding **conference proceedings**. All database searches were documented (query string, platform, and execution date). Results were deduplicated prior to screening.
Note on exclusions: Because some non-EEG wearable studies rely on video-EEG for labeling, EEG-related terms were used to filter **EEG-only modalities**, not to exclude studies that mention EEG solely as a reference standard.

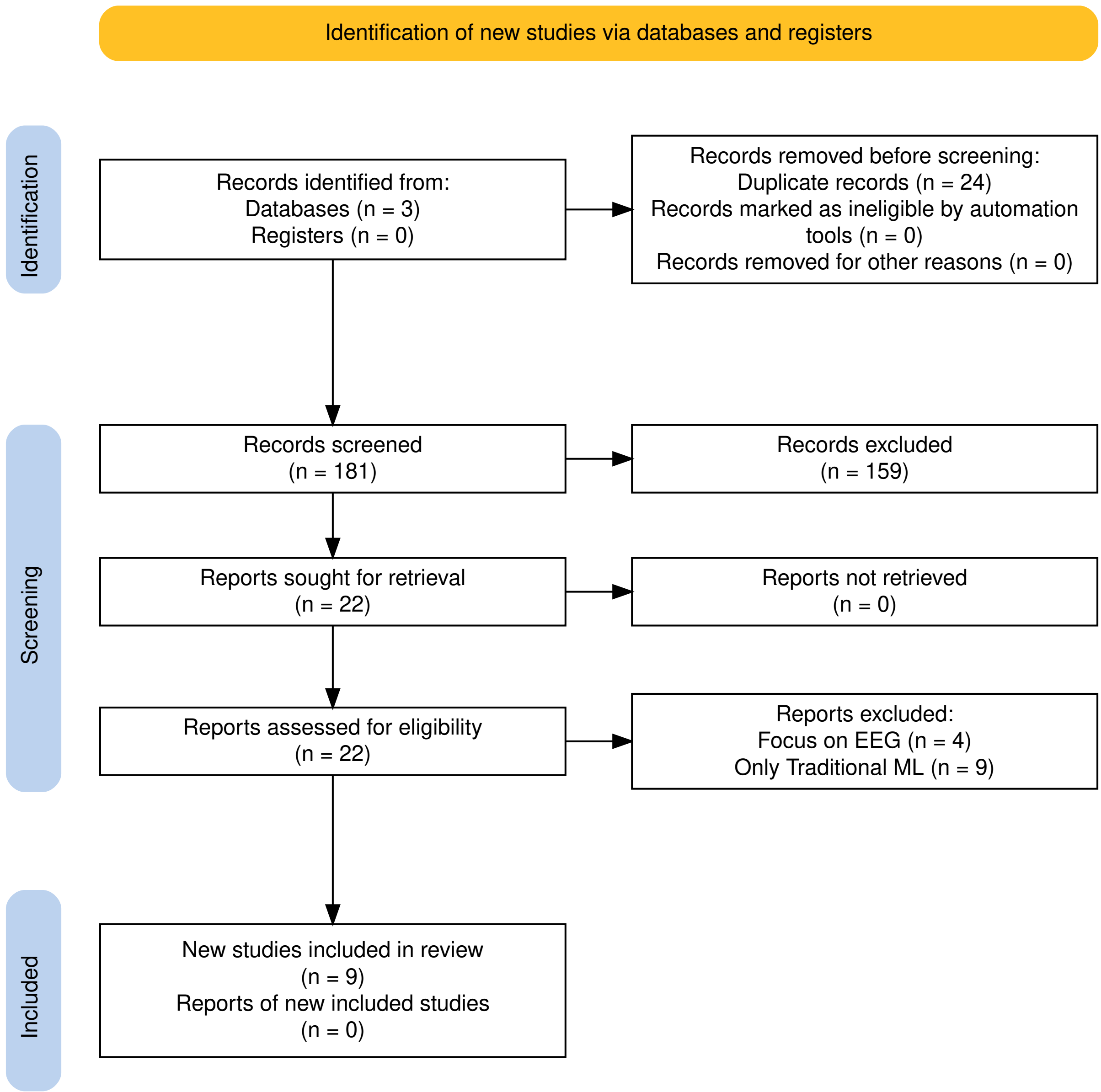

Figure 1: PRISMA flow diagram of the study selection process.

Table 1: Webster & Watson Concept Matrix: Detection Studies

Study	Design	Sample	Device	Loc.	Para.	Mod.	Pers.	Algorithm	Sens	Spec	FPR
Spahr et al. 2025	Prospective	n=384, 49 CSs	Empatica E4 (ACC)	Wrist	Ens	1	Glb	Ensemble 1D CNN (30 models)	96%	–	0.0054/h
Reintjes et al. 2025	Retro	n=120, 856 szs	Single-lead ECG	Chest	Anom	1	Subj	Anomaly: Matrix Profile, MADRID, TimeVQVAE	38.0%/98.16%	–	0.05–65.62/h
Fine 2025	Phase 1	n=15/3	6-axis band (ACC+Gyro)	Wrist	Feat	M	Glb	ANN (594 hand-crafted feat)	100%Test, 96% complete	–	0.023/h
Dong et al. 2026	Prospective	n=68, 1846 szs	NightWatch	Arm	2-stg	M	Glb	CNN-LSTM + Attention	71.6%	–	0.165/h
Wang et al. 2025	–	n=28, 62 szs	Biovital-P1	Wrist	Feat	M	Glb	LSTM (40 hidden)	56.40% to 95.3%	–	0.364/h
Singh Rathore 2024	Case study	1 pt, 6 hrs	Wearable (EDA+ACC+HR)	–	Feat	M	CS	MLP (multi-modal features)	96.8%	94.8%^{prec}	–
Elemam et al. 2025	Cross-sect	Q:198, HRV:30	Camera PPG	Thumbs	Hyb	M	Glb	CNN + Rule-based fusion	95.1% Q, 93% HRV	97.06% Q	–
Borujeny 2013	3-pt	3 pts, 20 szs	MICAz ACC	Arm/thigh	Feat	1	Glb	ANN (time-domain feat)	85%	–	3 FA

Note: ACC = accelerometer, ANN = artificial neural network, CNN = convolutional neural network, CS = convulsive seizure / case study, ECG = electrocardiogram, EDA = electrodermal activity, HR = heart rate, HRV = HR variability, LSTM = long short-term memory, PPG = photoplethysmography, Sens = sensitivity, Spec = specificity, sz(s) = seizure(s), FPR = false positive rate, – = not reported. Para. = learning paradigm (Feat = feature-based, Ens = ensemble, Anom = anomaly detection, 2-stg = two-stage, Hyb = hybrid). Mod. = modality count (1 = single-modal, M = multi-modal). Pers. = personalization (Glb = global model, Subj = subject split, CS = case study). ^{prec} = precision (not specificity).

Table 2: Webster & Watson Concept Matrix: Forecasting Studies

Study	Design	Sample	Device	Loc.	Para.	Mod.	Pers.	Algorithm	Sens	Spec	FPR
Vieluf et al. 2025	Retro	n=70, 5437 d	Embrace	Wrist	Hyb	M	Mix	DNN + harmonic features + diary	82%	67%	–
Meisel et al. 2020	LOSO CV	n=69, 452 szs	Empatica E4	Wrist/Ankle	E2E	M	Glb	LSTM (10 units)	51.2%	–	TiW 43.7%
Stirling 2021	Retro+pseudo	n=11, 136 szs	Fitbit	Wrist	Feat	M	Pat	LSTM+RF+LR ensemble (HRV feat)	–	–	AUC 0.74
Nasseri 2021	Retro	n=6	Empatica E4	Wrist	E2E	M	Tmp	LSTM 4-layer (128 hidden)	AUC 0.75 (SD 0.15)	–	TiW 0.9–7.2h/d
Ode et al. 2023	Retro	n=66, 85 szs	ECG	–	Anom	1	Pat	Self-Attentive AE	74%	–	0.85/h

Note: ACC = accelerometer, AE = autoencoder, CNN = convolutional neural network, DNN = deep neural network, ECG = electrocardiogram, EDA = electrodermal activity, LSTM = long short-term memory, LOSO = leave-one-subject-out, PPG = photoplethysmography, Sens = sensitivity, Spec = specificity, sz(s) = seizure(s), FPR = false positive rate, TiW = time in warning, AUC = area under curve, – = not reported. Para. = learning paradigm (Feat = feature-based, E2E = end-to-end, Anom = anomaly detection, Hyb = hybrid). Mod. = modality count (1 = single-modal, M = multi-modal). Pers. = personalization (Glb = global model, Pat = patient-specific, Mix = mixed, Tmp = temporal split).

Table 3: Architecture and Modality Deep-Dive: Detection Studies

Study	Paradigm	Arch. Details	Modality Details	Fusion	Personalization	Trade-off Config	Deployment
Spahr et al. 2025	Ensemble	30 1D CNN models, 14 conv layers, quantile	ACC 32 Hz, Euclidean norm, 30 s	–	Global, tunable	86–100% @ 0.0054/h	Real-time (112 ms), on-device
Reintjes et al. 2025	Anomaly	Matrix-Profile, MADRID, TimeVQVAE-AD	ECG 256→8 Hz, bandpass	–	Subject split	38.0–98.2% @ 0.05–65.62/h	Offline, hospital
Fine 2025	Feature-based	ANN, 594 handcrafted features	ACC+Gyro 6-axis, 10 s intervals	Early	Global, hold-out	100% @ 0.023/h	Offline PC, EMU
Dong et al. 2026	Two-stage	Pre-screening + CNN-LSTM + Attention	ACM 11-12→20 Hz, PPG 100→20 Hz, 5 min	Early	Global, 10-fold CV	71.6% @ 0.165/h	Real-time, NightWatch, home
Wang et al. 2025	Feature-based	LSTM (40 hidden) + ReLU + FC	ACC/GYR 50 Hz, SEMG 200 Hz, EDA 4 Hz, 4 s	Early	Global, hold-out	56.4–95.3% @ 0.364/h	Real-time, hospital
Singh 2024	Feature-based	MLP, multi-modal features	EDA, ACC, HR/HRV, 25k points	Early	Single pt (CS)	96.8% @ 94.8% prec	Real-time, cloud?
Elemam 2025	CNN-based	CNN for HRV, CNN for Audio (separate)	PPG 250 Hz, Audio 10 s	No fusion	Global, unclear	95.1% @ 97.06% spec	Real-time, hospital
Borujeny 2013	Feature-based	ANN, time-domain features	ACC 2D 3 Hz, 9 s	–	Global, 3 pts	85% @ 3 FA	Real-time, server (316 mW)

Note: Paradigm: Feature-based = handcrafted features, Ensemble = multiple models, Two-stage = sequential processing, Anomaly = unsupervised detection, CNN-based = convolutional approach. Arch. Details: architecture-specific components. Fusion: – = single-modal, Early = feature-level, No fusion = parallel separate models. Personalization: Global = population model, Subject split = train/test separated by subject. Trade-off Config: operating points with sensitivity @ FAR. Deployment: real-time capability, processing location, setting. ACC = accelerometer, ECG = electrocardiogram, PPG = photoplethysmography, EDA = electrodermal activity, HR = heart rate, HRV = HR variability, SEMG = surface electromyography, EMU = epilepsy monitoring unit.

Table 4: Architecture and Modality Deep-Dive: Forecasting Studies

Study	Paradigm	Arch. Details	Modality Details	Fusion	Personalization	Trade-off Config	Deployment
Vieluf et al. 2025	Hybrid	DNN + harmonic modeling (24-h)	EDA 4 Hz, ACC 32 Hz, Temp 1 Hz, 10 min	Early	Mixed, 5-fold + LOO	82% sens, 67% spec @ F1 0.81	Offline MATLAB, home
Meisel et al. 2020	End-to-end	LSTM only (10 units)	EDA/ACC/BVP/Temp $\rightarrow$4 Hz, 30 s	Early	Global, LOSO	51.2% @ IoC 14.1%	Offline, hospital
Stirling 2021	Feature-based	LSTM+RF+LR ensemble, LR combines	HR 5 s, steps/sleep 1 min, hourly/daily	Decision	Patient-specific, weekly	AUC 0.74, 100% pts	Home (Fitbit), app
Nasseri 2021	End-to-end	LSTM 4-layer, 128 hidden + FFT channels	ACC/BVP/EDA/Temp/HR $\rightarrow$128 Hz, 60 s	Early (17-ch)	Temporal split	AUC 0.75 (SD 0.15) @ TiW 0.9–7.2h/d	Offline, cloud, home
Ode et al. 2023	Anomaly	Self-Attentive AE (attention)	ECG (RRI only), 45 s	–	Patient-specific (99% CL)	74% @ 0.85/h (99% CL)	Real-time, cloud, hospital

Note: Paradigm: Feature-based = handcrafted features, End-to-end = learns from raw data, Hybrid = combines both, Anomaly = unsupervised detection. Arch. Details: architecture-specific components. Fusion: – = single-modal, Early = feature-level, Decision = classifier output. Personalization: Global = population model, Patient-specific = individualized, Temporal = time-split within patient, LOSO = leave-one-subject-out, LOO = leave-one-out, Mixed = combination of approaches. Trade-off Config: operating points with sensitivity @ FAR. Deployment: real-time capability, processing location, setting. ACC = accelerometer, ECG = electrocardiogram, EDA = electrodermal activity, BVP = blood volume pulse, HR = heart rate, TEMP = temperature, FFT = fast Fourier transform, CL = confidence limit, AUC = area under curve, TiW = time in warning, IoC = Improvement over Chance.

Table 5: Performance Metrics Reporting Across Studies

Metric	Detection (n=9)	Forecasting (n=4)	Reported By
Sensitivity/Recall	8/8	1/4	Spahr, Reintjes, Fine, Dong, Wang, Singh Rathore, Elemam, Borujeny, Ode[a]
Specificity	3/9	0/4	Elemam, Vieluf, Wang
FPR/FA rate	8/9	0/4	Spahr, Reintjes, Fine, Dong, Wang, Singh Rathore, Elemam, Borujeny, Nasseri, Ode
AUC-ROC	3/9	3/4	Dong, Reintjes, Ode, Nasseri, Stirling
Detection Latency	4/9	1/4	Spahr, Fine, Elemam, Borujeny, Nasseri
Patient Success Rate	2/9	3/4	Reintjes, Meisel, Stirling, Nasseri, Ode
Precision/PPV	4/9	1/4	Dong, Singh Rathore, Elemam, Ode
IoC / Time in Warning	0/9	3/4	Meisel, Stirling, Nasseri

5 Results

This systematic review analyzed 13 primary studies published between 2013 and 2026. The evidence base comprises 9 detection studies and 4 forecasting studies with a total of 912 participants. Table 1 and Table 2 present the study matrices for detection and forecasting approaches, respectively. Table 3 and Table 4 provide architectural details, and Table 5 summarizes the metrics reported across studies.

5.1 Study Characteristics

The 13 studies included in this review exhibit substantial variation in sample size and study design. Detection studies enrolled a median of 44 participants (range 1 to 384), while forecasting studies enrolled a median of 11 participants (range 6 to 70). The total sample comprises 687 participants in detection studies and 225 in forecasting studies.

Three studies employed prospective designs. Spahr et al. (2025) conducted a multi-center prospective study across eight epilepsy monitoring units with 384 patients. Dong et al. (2022) prospectively monitored 68 patients at home for up to three months. Fine (2025) prospectively recorded data from 15 patients in an epilepsy monitoring unit for tonic seizure detection. The remaining studies used retrospective analysis of existing datasets or small-scale feasibility designs.

Research activity has accelerated over time. Only one study was published between 2013 and 2015. Three forecasting studies were published between 2020 and 2022. Nine detection studies were published between 2023 and 2026. This pattern suggests increased research attention on detection approaches in recent years.

Eight studies were conducted in hospital or EMU settings. Six studies included home or ambulatory validation. Dong et al. (2022) achieved the longest home monitoring duration with 788 overnight recordings over three months. Stirling et al. (2021) achieved the longest overall monitoring period with a mean of 14.6 months per patient using consumer smartwatches.

Key limitation. The evidence base includes four studies with sample sizes below 20 participants. Singh Rathore et al. (2024) is a single-patient case study with limited generalizability. Borujeny et al. (2013) evaluated only 3 patients. Small sample sizes limit the reliability of performance estimates and generalizability to broader populations.

5.2 Modalities, Architectures, and Personalization

5.2.1 Modality Selection and Sensor Placement

Biosignal modality selection differs between detection and forecasting approaches. Detection studies prioritize movement sensors, while forecasting studies emphasize autonomic nervous system indicators. Accelerometer data appears in nine of 13 studies (69%), with six detection studies and three forecasting studies using accelerometry. Electrodermal activity appears in five studies (38%). ECG or HRV data appears in five studies. PPG appears in four studies. Gyroscope data appears in three studies, all in detection applications.

Multi-modal approaches dominate the evidence base. Nine of 13 studies (69%) use multiple sensor types. Four studies use single-modality approaches: Spahr et al. (2025) (ACC only), Reintjes et al. (2025) (ECG only), Borujeny et al. (2013) (ACC only), and Ode et al. (2023) (ECG RRI only). Wrist-worn devices appear in eight studies (62%), reflecting patient acceptance and the availability of consumer wearable devices.

Single-modality accelerometer approaches achieve strong performance. Spahr et al. (2025) reported 96% sensitivity with 0.0054/h FPR using only ACC data, representing the lowest FPR among all detection studies. Multi-modal movement detection shows excellent sensitivity in controlled settings, with Fine (2025) achieving 100% sensitivity using a 6-axis band combining ACC and gyroscope. ECG-based detection shows high FPR variability, with Reintjes et al. (2025) reporting FPR ranging from 1.91/h to 39.75/h depending on the anomaly detection method.

Forecasting studies consistently use autonomic signals. Nasseri et al. (2021) combined ACC, PPG, EDA, temperature, and HR to achieve AUC 0.80 (SD 0.15). Meisel et al. (2020) used EDA, PPG, temperature, and ACC to achieve 51.2% sensitivity with IoC 14.1% in LOSO validation. AUC plateaus around 0.75–0.80 regardless of modality combination in forecasting applications.

5.2.2 Architecture Patterns

Deep learning architectures differ substantially between detection and forecasting applications. Detection research employs diverse paradigms, while forecasting studies concentrate on LSTM-based approaches.

CNN-based approaches dominate detection research. Spahr et al. (2025) implemented an ensemble of 30 CNN models, achieving 96% sensitivity with 0.0054/h FPR. Elemam et al. (2025) used separate CNNs for HRV and audio processing. LSTM appears in one detection study, with Wang et al. (2025) using an LSTM with 40 hidden units processing attitude angle features. Traditional neural networks appear in earlier studies, with Fine (2025) using an ANN with 594 handcrafted features.

LSTM architectures dominate forecasting research. Three of five forecasting studies use LSTM variants. Meisel et al. (2020) implemented a minimal LSTM with 10 hidden units operating directly on raw multi-modal wrist data. Nasseri et al. (2021) used a deeper 4-layer LSTM with 128 hidden units per layer. Stirling et al. (2021) combined LSTM with random forest and logistic regression in an ensemble. No forecasting study uses CNN-based approaches, possibly because forecasting requires modeling temporal dependencies rather than spatial patterns.

Handcrafted features remain competitive. Three of eight detection studies use feature-based approaches with strong performance. Fine (2025) achieved 100% sensitivity using

594 handcrafted features. This suggests that domain knowledge remains valuable despite the end-to-end learning trend. Window sizes vary substantially between applications. Detection studies use shorter windows ranging from 4 seconds to 5 minutes. Forecasting studies use longer windows ranging from 30 seconds to 60 minutes.

5.2.3 Personalization Trade-offs

Personalization strategy represents a fundamental design choice. Global models train on population data and apply to all patients. Patient-specific models train on individual patient data. Five studies use global model approaches. Four studies use patient-specific approaches. Vieluf et al. (2025) used a mixed approach combining population-level harmonic patterns with individual diary data.

Global models offer advantages for deployment. They enable on-device processing without requiring individual patient training. Spahr et al. (2025) achieved 112 ms inference time suitable for real-time processing. However, global models may not capture individual seizure patterns. Meisel et al. (2020) used LOSO validation to assess global model generalizability, achieving 62% patient success (43 of 69 patients above chance).

Patient-specific models achieve higher individual success rates but require substantial individual data. Stirling et al. (2021) achieved 100% patient success for hourly forecasting with weekly retraining. Nasseri et al. (2021) achieved 83% patient success (5 of 6 patients). These rates are substantially higher than the 62% achieved by global models. However, patient-specific approaches require a mean of 14.6 months of data per patient or 60+ days per patient. This data requirement creates a cold start problem where new patients must wait weeks or months before the system becomes useful.

The trade-off is clear. Patient-specific models achieve higher individual success (83–100%) compared to global models with LOSO validation (62%), but they require weeks to months of individual data collection. Global models enable immediate deployment and on-device processing but may not work well for all patients. Mixed approaches may offer a middle path, though evidence remains limited to a single study.

5.3 Performance Metrics

Detection and forecasting studies use different evaluation metrics reflecting their distinct clinical objectives. Detection studies emphasize sensitivity and false positive rate (FPR). Forecasting studies report AUC-ROC, improvement over chance (IoC), and time in warning (TiW).

5.3.1 Detection Performance

Detection sensitivity ranges from 38.0% to 100% across studies (Table 5). Fine (2025) reported 100% sensitivity in an independent test set of 10 tonic seizures from 3 patients. Spahr et al. (2025) achieved 96% sensitivity for generalized convulsive seizures in a 384-patient multi-center study. Singh Rathore et al. (2024) reported 96.8% sensitivity in a single-patient case study. Elemam et al. (2025) achieved 95.1% sensitivity using HRV-based detection.

Reintjes et al. (2025) reported the widest sensitivity range from 38.0% to 98.2% depending on the anomaly detection method. The TimeVQVAE-AD method achieved 98.2% sensitivity, while the Matrix Profile method achieved only 38.0%.

Only six of nine detection studies report FPR. The ILAE Phase 3 benchmark for clinically acceptable FPR in home settings is below 0.05 to 0.1 per hour. Two studies meet this benchmark. Spahr et al. (2025) reported 0.0054/h FPR (approximately 1 false alarm per 8 days), which is approximately 10 times below the clinical benchmark. Fine (2025) reported 0.023/h FPR, approximately twice below the benchmark.

Three studies exceed the clinical benchmark. Dong et al. (2022) reported 0.165/h FPR. Wang et al. (2025) reported 0.364/h FPR. Ode et al. (2023) reported 0.85/h FPR, which is 8.5 times above the benchmark. Reintjes et al. (2025) reported FPR ranging from 0.05/h to 65.62/h depending on method and configuration, which is clinically unacceptable for home use.

Three detection studies do not report FPR. Singh Rathore et al. (2024), Elemam et al. (2025), and Borujeny et al. (2013) lack standardized FPR reporting. This omission complicates clinical assessment and comparison across studies.

5.3.2 Forecasting Performance

Forecasting studies use different metrics than detection studies. AUC-ROC serves as the primary metric in two forecasting studies. Nasseri et al. (2021) reported AUC 0.75 (SD 0.15) across 6 patients with temporal split validation. Stirling et al. (2021) reported AUC 0.74 with 100% patient success.

Sensitivity in forecasting ranges from 51.2% to 82%. Vieluf et al. (2025) reported 82% sensitivity with 67% specificity. Meisel et al. (2020) reported 51.2% sensitivity with LOSO validation.

Forecasting studies also report forecasting-specific metrics. Meisel et al. (2020) reported IoC of 14.1% and TiW of 43.7%. Nasseri et al. (2021) reported TiW ranging from 0.9 to 7.2 h/day. Stirling et al. (2021) reported mean prediction time of 37 min for hourly forecasts and 3 days for daily forecasts.

Forecasting AUC consistently falls between 0.74 and 0.75 across studies. This consistency suggests a performance ceiling for non-invasive forecasting approaches using wearable biosignals.

5.4 Detection, Forecasting, and Clinical Readiness

Detection and forecasting serve different clinical purposes. Detection aims to identify seizures as they occur for timely intervention, while forecasting aims to predict seizures before onset to allow preventive action. These different objectives lead to distinct technical approaches, validation requirements, and clinical readiness levels.

5.4.1 Maturity Comparison

Detection shows greater clinical maturity than forecasting. Two detection studies meet Phase 3 FPR benchmarks defined by the ILAE, though both are Phase 2 studies themselves. Spahr et al. (2025) achieved 0.0054/h FPR (approximately 1 false alarm per 8 days) with 96% sensitivity in a prospective multi-center study of 384 patients. In a commentary on Larsen et al. 2024, Fine (2025) described 0.023/h FPR with 100% sensitivity on an independent test set of 3 patients with 10 tonic seizures. Commercial devices exist for detection including the Embrace watch with FDA 510(k) clearance and NightWatch with CE approval in Europe.

Forecasting shows consistent AUC around 0.74 to 0.75 across studies, suggesting a performance ceiling for non-invasive approaches. Meisel et al. (2020) reported AUC 0.74 with IoC of 14.1% over chance. Nasseri et al. (2021) reported AUC 0.80 (mean) but studied only 6 patients. Vieluf et al. (2025) achieved AUC 0.75 in a larger sample of 70 patients. However, no forecasting study meets a clear clinical benchmark and no device has regulatory approval for seizure forecasting.

Detection benefits from clearer clinical requirements. The ILAE has defined guidelines for detection device validation including FPR targets below 0.05 to 0.1 per hour. Forecasting lacks equivalent standardized guidelines. This gap complicates assessment of forecasting readiness and clinical utility.

Sample size disparities further reflect the maturity difference. Detection studies have a median sample size of 66 participants compared to 40 for forecasting studies. Larger detection studies enable more robust performance estimates. The largest detection study enrolled 384 patients across eight centers, while the largest forecasting study enrolled 70 patients.

5.4.2 Barriers to Clinical Adoption

Several barriers impede clinical translation of wearable seizure monitoring. First, high FPR limits clinical utility. Six of eight detection studies report FPR above acceptable thresholds. Dong et al. (2022) reported 0.165/h FPR in home validation, Wang et al. (2025) reported 0.354/h, and Reintjes et al. (2025) reported FPR from 1.91/h to 39.75/h. High FPR causes alarm fatigue and may lead patients to abandon devices.

Second, limited seizure type coverage restricts applicability. Detection approaches focus on convulsive seizures with motor manifestations captured by accelerometer and gyroscope sensors. Focal aware seizures without motor signs remain undetectable with current wearable approaches. Non-convulsive seizures represent a substantial unmet need that may require invasive EEG monitoring.

Third, small sample sizes limit generalizability. Four studies have sample sizes below 20 patients. Singh Rathore et al. (2024) is a single-patient case study. Borujeny et al. (2013) studied only 3 patients. Small samples increase uncertainty about performance estimates and may not capture the full diversity of seizure presentations.

Fourth, validation setting affects real-world applicability. Seven studies were conducted entirely in hospital or EMU settings. Hospital environments may not capture the diversity of real-world activities that cause false alarms. Exercise, cooking, household chores, and other daily activities present challenges not encountered in controlled environments.

Fifth, device burden and battery life limit adherence. Multi-modal systems sampling multiple signals at high frequencies consume substantial power. Long-term monitoring requires daily charging, which may reduce adherence over time. Stirling et al. (2021) achieved the longest monitoring duration with a mean of 14.6 months per patient, but such extended use requires careful attention to device comfort and reliability.

5.4.3 Most Advanced Approaches

Spahr et al. (2025) represents the most clinically advanced detection approach. The study combines large sample size, prospective multi-center design, excellent sensitivity, and the lowest reported FPR among detection studies. The ensemble 1D CNN algorithm operates in real-time with 112 ms inference time suitable for on-device processing. The main limitation is EMU setting rather than home validation.

Dong et al. (2022) represents the most advanced home-validated detection approach. The study achieved prospective home monitoring with 788 overnight recordings over three months using the NightWatch commercial device. FPR of 0.165/h exceeds clinical benchmarks but may be acceptable for some patients prioritizing sensitivity over specificity.
Stirling et al. (2021) represents the most clinically advanced forecasting approach. The study achieved 100% patient success with long home validation using consumer Fitbit devices. The LSTM+RF+LR ensemble with weekly retraining adapts to individual patient patterns. However, forecasting lacks clear clinical benchmarks and regulatory pathways.
Key limitation. No study has achieved all requirements for clinical deployment. All studies have limitations in sample size, validation setting, FPR, or generalizability. No study has achieved Phase 3 prospective validation in home settings with sufficient sample size and duration. Substantial additional research is needed before wearable seizure monitoring can achieve widespread clinical adoption.

6 Discussion

This discussion synthesizes findings from 13 studies on wearable seizure detection and forecasting. The evidence reveals different maturity levels for detection versus forecasting, distinct technical requirements, and persistent barriers to clinical translation.

6.1 Clinical Implications

6.1.1 Detection Readiness

Two detection studies meet the ILAE Phase 3 benchmark for clinically acceptable false positive rates. Spahr et al. (2025) reported 0.0054/h FPR, approximately 10 times below the 0.05 to 0.1/h threshold. Fine (2025) reported 0.023/h FPR, approximately twice below the threshold. Both studies used ACC-based approaches, suggesting this modality has strong translation potential.
Accelerometer-based detection shows particular promise for clinical deployment. Six of nine detection studies use ACC data. The motor manifestations of convulsive seizures produce clear signals in ACC recordings. This biological specificity enables high sensitivity with low FPR when combined with deep learning approaches.
A significant gap remains between EMU validation and home deployment. Only three detection studies include home validation. The two studies meeting FPR benchmarks were conducted in EMU settings. Spahr et al. (2025) enrolled 384 patients across eight centers but tested in hospital environments. Fine (2025) tested only 10 seizures from 3 patients. Home environments present challenges not encountered in hospital settings, including exercise, household chores, and other activities that may cause false alarms.
Dong et al. (2022) achieved the most extensive home validation with 788 overnight recordings over three months. However, the 0.165/h FPR exceeds clinical benchmarks. This study demonstrates both the feasibility of home monitoring and the challenge of maintaining low FPR in real-world conditions.
Commercial devices exist for detection but face limitations. The Embrace watch has FDA 510(k) clearance for seizure detection. NightWatch has CE approval in Europe. However, these approvals were based on earlier validation studies. The deep learning approaches reviewed here may offer improved performance but require separate validation pathways.

6.1.2 Forecasting Status

Forecasting shows a consistent performance ceiling across studies. Three forecasting studies report AUC between 0.74 and 0.75. Meisel et al. (2020) reported AUC 0.74. Nasseri et al. (2021) reported mean AUC 0.80 for individual models but 0.75 for the ensemble. Stirling et al. (2021) reported AUC 0.74. This consistency across different methods, modalities, and patient populations suggests a fundamental limit of non-invasive forecasting approaches.

No established clinical benchmarks exist for forecasting performance. The ILAE has defined guidelines for detection device validation but not for forecasting. IoC of 14.1% (reported by Meisel et al. (2020)) suggests modest improvement over chance. However, the clinical acceptability of this performance level remains unclear. TiW reported by forecasting studies indicates patients spend substantial time in warning states, which may affect quality of life.

Patient success rates vary substantially by personalization strategy. Global LOSO validation achieved 62% patient success (Meisel et al. (2020)). Patient-specific weekly retraining achieved 100% patient success (Stirling et al. (2021)). This trade-off between generalizability and individual performance remains unresolved for clinical deployment.

No device has regulatory approval specifically for seizure forecasting. Forecasting devices would need to establish clinical utility and safety through dedicated trials. The lack of clear benchmarks complicates pathway to regulatory approval.

6.1.3 Seizure Type Coverage Gap

Most approaches detect only convulsive or motor seizures. Detection studies emphasize movement sensors that capture motor manifestations. Six of nine detection studies use ACC data. Three use gyroscope data. These modalities cannot detect seizures without motor components.

Non-convulsive seizures represent a substantial unmet need. Focal aware seizures without motor signs remain undetectable with current wearable approaches. Reintjes et al. (2025) showed that ECG-based detection can identify some non-motor seizures through autonomic changes. However, sensitivity for non-motor focal aware seizures reached only 51.1% compared to 96.7% for focal-to-bilateral tonic-clonic seizures.

Forecasting studies emphasize autonomic modalities rather than movement sensors. Four of five forecasting studies use EDA. Three use ECG or HRV data. These modalities may capture pre-ictal autonomic changes that precede both convulsive and non-convulsive seizures. However, the consistent AUC ceiling around 0.75 suggests fundamental limitations in predicting seizures without EEG biomarkers.

Approximately 30% of people with epilepsy have seizures that would not be detected by current wearable approaches. Non-convulsive focal seizures, absence seizures, and seizures with minimal autonomic changes require alternative monitoring strategies. Invasive EEG remains the gold standard for comprehensive seizure monitoring but is not suitable for long-term ambulatory use.

6.2 Technical Insights and Trade-offs

6.2.1 Modality Paradox

Multi-modal approaches dominate wearable seizure monitoring research. Nine of 13 studies (69%) use multiple sensor types. However, multi-modal does not automatically mean superior performance. The study with the lowest false positive rate in the detection literature uses single-modality accelerometry. Spahr et al. (2025) achieved 0.0054/h FPR with ACC only, well below the clinical benchmark of 0.05–0.1/h. This finding suggests that sensor fusion is not always necessary for reliable detection.

The contrast between detection and forecasting applications reveals different modality requirements. Detection studies can succeed with movement sensors alone. Forecasting studies consistently require autonomic modalities. All four forecasting studies incorporate EDA, ECG, or temperature data. Vieluf et al. (2025) demonstrated that 24-hour harmonic patterns in EDA and accelerometry contain predictive information for seizures. This pattern reflects the different physiological targets. Detection captures ictal motor manifestations. Forecasting must detect pre-ictal autonomic changes that precede clinical symptoms.

ECG-based detection shows particular challenges. Reintjes et al. (2025) compared three anomaly detection methods on single-lead ECG. FPR ranged from 1.91/h to 39.75/h, far exceeding clinical acceptability. Even the best ECG-based approach by Ode et al. (2023) produced 0.85/h FPR. These results suggest that cardiac alone may be insufficient for reliable detection.

6.2.2 Architecture Selection

Detection and forecasting applications employ different architectural patterns. Detection studies show methodological diversity including CNNs, LSTMs, ANNs, ensemble methods, and anomaly detection. Forecasting studies concentrate on LSTM variants. Three of four forecasting studies use LSTM architectures. This pattern reflects the different temporal demands of the two tasks. Detection requires identifying patterns within short windows. Forecasting requires modeling longer-term temporal dependencies.

Handcrafted features remain competitive despite the trend toward end-to-end learning. Fine (2025) achieved 100% sensitivity with 594 handcrafted features from 6-axis ACC and gyroscope data. Wang et al. (2025) found that engineered attitude angle features outperformed raw sensor data. These results suggest that domain knowledge still adds value in seizure monitoring applications.

Ensemble methods show promise for detection. Spahr et al. (2025) trained 30 separate 1D CNN models with quantile aggregation. This approach achieved the best FPR in the detection literature at 0.0054/h. However, ensemble methods require greater computational resources. The single-modality ACC ensemble achieved 112 ms inference time, still suitable for real-time processing.

6.2.3 Personalization Dilemma

Personalization strategy involves a fundamental trade-off between deployability and individual performance. Global models enable immediate deployment without individual patient training. Patient-specific models achieve higher individual success rates but require substantial individual data.

The contrast in patient success rates is striking. Meisel et al. (2020) used LOSO validation to assess global model generalizability. Only 62% of patients (43 of 69) achieved performance above chance. By contrast, Stirling et al. (2021) achieved 100% patient success with patient-specific models. This pattern suggests that population models may not work well for a substantial minority of patients.

The cold start problem limits patient-specific approaches. Stirling et al. (2021) required a mean of 14.6 months of data per patient before models became useful. Nasseri et al. (2021) required 60+ days per patient. These data requirements mean new patients must wait weeks or months before the system provides protection.

Mixed approaches may offer a middle path. Vieluf et al. (2025) combined population-level 24-hour harmonic patterns with individual diary data. This approach achieved 82% sensitivity and 67% specificity. Further research on hybrid personalization strategies could balance immediate utility with individual adaptation.

6.2.4 The False Positive Challenge

False alarm rate remains the primary barrier to clinical translation. Six of eight detection studies exceed the clinical FPR benchmark of 0.05–0.1/h. Only two studies achieve acceptable FPR for home use. Spahr et al. (2025) reported 0.0054/h FPR with single-modality ACC. Fine (2025) reported 0.023/h FPR with a 6-axis band.

The problem is particularly acute for ECG-based approaches. Reintjes et al. (2025) found that anomaly detection methods produced FPR ranging from 1.91/h to 39.75/h depending on the algorithm. Even the best performing method exceeded the clinical benchmark by a factor of 19. These results suggest that cardiac signals alone cannot reliably distinguish seizures from normal activity.

Multi-modal approaches do not guarantee acceptable FPR. Dong et al. (2022) combined ACC and PPG in a two-stage architecture but achieved 0.165/h FPR, still above clinical targets. Wang et al. (2025) combined four modalities but reported 0.364/h FPR. Sensor fusion alone does not solve the false positive problem.

The forecasting challenge differs. Forecasting studies report AUC rather than FPR. The best forecasting studies plateau around AUC 0.75. Nasseri et al. (2021) achieved AUC 0.80 (SD 0.15) with patient-specific models. Stirling et al. (2021) achieved AUC 0.74 with heart rate cycles. This performance ceiling suggests fundamental limits on pre-ictal predictability using wearable sensors.

6.3 Limitations of the Evidence Base

6.3.1 Sample Size Constraints

The evidence base for wearable seizure monitoring is constrained by small sample sizes in multiple studies. Detection studies enrolled a median of 44 participants, while forecasting studies enrolled a median of 11 participants. Four studies have sample sizes below 20 participants. Singh Rathore et al. (2024) is a single-patient case study with limited generalizability. Borujeny et al. (2013) evaluated only 3 patients. Fine (2025) tested 10 seizures from 3 patients. Small sample sizes limit the reliability of performance estimates and reduce confidence that results will generalize to broader epilepsy populations.

Forecasting studies face particular sample size challenges. Nasseri et al. (2021) studied only 6 patients due to the requirement for invasive EEG confirmation via the RNS system. Stirling et al. (2021) studied 11 patients. Small samples increase uncertainty about performance estimates and may not capture the heterogeneity of seizure patterns across patients.

6.3.2 Validation Setting Bias

Eight of 13 studies were conducted in hospital or EMU settings. Only six studies included home or ambulatory validation. This setting bias limits the applicability of findings to real-world use. Hospital environments lack the diversity of activities that generate false alarms in daily life. Exercise, cooking, household chores, and other routine activities present motion patterns and physiological changes not encountered in controlled hospital environments.

Spahr et al. (2025) achieved excellent performance with 0.0054/h FPR in an EMU setting. However, this performance may not translate to home use where patients engage in varied activities. Dong et al. (2022) conducted home monitoring and reported 0.165/h FPR, substantially higher than the EMU-based studies. This discrepancy suggests that hospital-based validation may underestimate real-world FPR.

Forecasting studies show stronger home validation. Stirling et al. (2021) achieved a mean of 14.6 months monitoring per patient using consumer smartwatches. Nasseri et al. (2021) achieved concurrent EEG confirmation via the RNS system in ambulatory settings. However, these studies remain limited by small sample sizes.

6.3.3 Reporting Inconsistency

Inconsistent metrics reporting hinders comparison across studies. Forecasting studies routinely report patient-level success rates. Stirling et al. (2021) achieved 100% patient success. Nasseri et al. (2021) achieved 83% patient success. Meisel et al. (2020) achieved 62% patient success with LOSO validation.

Detection studies rarely report patient-level results. Only two of nine detection studies provide individual patient outcomes. This reporting gap prevents assessment of which patients benefit from global detection models. High aggregate sensitivity may mask substantial variability across patients. Without patient-level reporting, it is unclear whether detection algorithms work reliably for all patients or only a subset.

6.3.4 Monitoring Duration and Cold-Start Problems

Most studies report limited monitoring durations. This constraint is particularly problematic for patient-specific approaches that require sufficient individual data for model training. Stirling et al. (2021) required a mean of 14.6 months of data per patient to achieve reliable forecasting. Nasseri et al. (2021) required 60+ days per patient.

This data requirement creates a cold start problem where new patients must wait weeks or months before the system becomes useful. Global models address this limitation by enabling immediate deployment. However, global models may not capture individual seizure patterns and achieve lower patient success rates. The trade-off between immediate deployment and individualized performance remains unresolved.

Dong et al. (2022) achieved the most extensive home validation with 788 overnight recordings over three months. This duration approaches clinically meaningful monitoring periods but remains insufficient for long-term assessment of device adherence and performance stability.

6.4 Future Directions

Four research priorities emerge from this review. First, forecasting needs standardized clinical benchmarks. Detection has the ILAE Phase 3 benchmark of FPR below 0.05

to 0.1 per hour. Forecasting lacks an equivalent standard. AUC values around 0.74 to 0.75 appear consistently across forecasting studies, but the clinical meaning of this performance level remains unclear. Metrics such as IoC and TiW need standardization. The field should establish minimum acceptable performance levels for forecasting devices to enable clinical translation.

Second, large-scale home validation is needed. Only two detection studies include prospective home validation. Hospital and EMU settings may not capture the diversity of real-world activities that cause false alarms. Exercise, cooking, household chores, and other daily activities present challenges not encountered in controlled environments. The 0.165/h FPR reported by Dong et al. (2022) in home settings exceeds the clinical benchmark. Future studies should prioritize prospective validation in intended use environments with sufficient sample sizes and duration.

Third, novel modalities are needed for non-convulsive seizures. Current approaches detect only motor seizures with convulsive manifestations. Focal aware seizures without motor signs remain undetectable with current wearable approaches. Spahr et al. (2025) achieved excellent performance for convulsive seizures using accelerometer data alone, but this approach cannot detect seizures without movement. Multi-modal autonomic approaches show promise but have not addressed non-convulsive seizure types. Novel biosignals or advanced feature engineering may be required to detect the subtle physiological changes associated with focal seizures.

Fourth, adaptive personalization strategies could address the trade-off between deployability and individual performance. Global models enable immediate deployment but may not work equally well for all patients. Meisel et al. (2020) achieved 62% patient success with LOSO validation, indicating substantial inter-patient variability. Patient-specific models achieve higher individual success but require weeks to months of individual data. Stirling et al. (2021) required a mean of 14.6 months per patient. Mixed approaches that combine population-level patterns with individual adaptation, such as the method used by Vieluf et al. (2025), may offer a balance. Research should address the cold-start problem by developing models that can initialize from population data and adapt efficiently to individual patients.

7 Conclusion

This systematic review analyzed 13 studies on wearable seizure detection and forecasting using deep learning approaches. The evidence base comprises 912 participants across nine detection studies and four forecasting studies published between 2013 and 2026. Three key findings emerge from this analysis.

First, detection approaches are approaching clinical readiness. Two studies meet the ILAE Phase 3 benchmark for false positive rates below 0.05 to 0.1 per hour. Spahr et al. (2025) achieved 0.0054/h FPR using an ensemble of 30 CNN models with single-modality ACC data. Fine (2025) achieved 0.023/h FPR using a 6-axis ACC and gyroscope band. Accelerometer-based detection shows strong translation potential for convulsive seizures. However, a significant gap remains between EMU validation and home deployment. Only three detection studies include home validation, and FPR increases substantially in real-world settings.

Second, forecasting approaches face a consistent performance ceiling. AUC values plateau around 0.74 to 0.75 across studies using different methods, modalities, and patient populations. Meisel et al. (2020) achieved AUC 0.74 with LOSO validation. Nasseri et al.

(2021) achieved AUC 0.80 with patient-specific models. Stirling et al. (2021) achieved AUC 0.74 with heart rate cycles. This consistency suggests fundamental limits on preictal predictability using non-invasive wearable sensors. Unlike detection, forecasting lacks established clinical benchmarks and regulatory pathways.

Third, three barriers impede clinical translation. The personalization dilemma remains unresolved. Global models enable immediate deployment but achieve only 62% patient success in LOSO validation. Patient-specific models achieve higher success rates but require weeks to months of individual data, creating a cold start problem. Seizure type coverage is limited to convulsive seizures with motor manifestations. Approximately 30% of people with epilepsy have non-convulsive seizures that remain undetectable with current wearable approaches. Real-world deployment presents challenges not captured in hospital validation, as evidenced by higher FPR in home settings.

No study in this review has achieved all requirements for widespread clinical adoption. Commercial devices exist for detection including the Embrace watch and NightWatch armband, but no device has regulatory approval for forecasting. Future research should prioritize home validation at scale, standardized benchmarks for forecasting performance, and novel modalities for non-convulsive seizure detection. Mixed personalization approaches that combine population patterns with individual adaptation may offer a path forward.

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